



Revisiting the expectations hypothesis of the term structure of interest rates

George Bulkley^a, Richard D.F. Harris^{b,*}, Vivekanand Nawosah^c

^a Department of Accounting and Finance, University of Bristol, 8 Woodland Road, Bristol BS8 1TN, UK

^b Xfi Centre for Finance and Investment, University of Exeter, Streatham Court, Rennes Drive, Exeter EX4 4ST, UK

^c Essex Business School, University of Essex, Wivenhoe Park, Colchester CO4 3SQ, UK

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ABSTRACT

The expectations hypothesis of the term structure has been decisively rejected in a large empirical literature that spans several decades. In this paper, using a newly constructed dataset of synthetic zero-coupon bond yields, we show that evidence against the expectations hypothesis is substantially weaker in data generated after the widespread publicity of its failure. These results are consistent with the idea that asset pricing anomalies tend to disappear once they are widely recognized.

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1. Introduction

The expectations hypothesis of the term structure of interest rates states that the yield on a long bond is equal to the average expectation of the short yield over the life of the long bond plus a constant risk premium. The expectations hypothesis (henceforth EH) has a number of important implications for the relationships between bond yields and their co-movement over time, and is one of the most widely tested theories in financial economics. The implications of the EH for the movement of bond yields have been investigated using a wide range of tests in a literature that spans several decades. The early tests of the EH examine the predictive ability of forward yields that are implicit in the term structure, and find that although informative, forward yields are biased predictors of future interest rates (Fama, 1984; Fama and Bliss, 1987). Campbell and Shiller (1991) develop a range of tests based on the yield spread between bonds of different maturities. Using both single equation and VAR-based approaches, they report overwhelming evidence against the EH. In many cases, bond yields

appear to move in a direction that is opposite to that predicted by theory.¹ Subsequent studies have shown that this rejection is not confined to the US.²

A natural question is whether the empirical failure of the EH can be accounted for within the rational expectations paradigm, or whether it constitutes evidence of an asset pricing anomaly. To this end, there has been a sustained search for a 'rational' explanation for the rejection of the EH. A limitation of standard tests of the EH is that they assume that the risk premium is constant. In the presence of a time-varying risk premium, tests of the EH are potentially biased in favor of its rejection. A number of studies have explored this possibility and indeed tests that allow for a time-varying risk premium have generally produced weaker rejections of the EH (see, for example, Fama, 1984; Evans and Lewis, 1994; Mankiw and Miron, 1986). However, the results of these studies are sensitive to the choice of proxy for the risk premium, the bond maturities considered and the sample period used.³ On balance, it

¹ See also Shiller (1979), Shiller et al. (1983), Campbell and Shiller (1984), Mankiw and Summers (1984), Mankiw (1986), Campbell (1995), Brown et al. (2008) and Beechey et al. (2009).

² See, for example, Hardouvelis (1994).

³ See also Shiller et al. (1983), Jones and Roley (1983), Backus et al. (1989), Simon (1989), Froot (1989), Tzavalis and Wickens (1997), Harris (2001), Cochrane and Piazzesi (2005), Kessler and Scherer (2009), Wright and Zhou (2009) and Sekkel (2011).

* Corresponding author. Tel.: +44 139 272 3215; fax: +44 139 226 2475.

E-mail addresses: George.Bulkley@bristol.ac.uk (G. Bulkley), R.D.F.Harris@exeter.ac.uk (R.D.F. Harris), vnawo@essex.ac.uk (V. Nawosah).

would appear that while the presence of a time-varying risk premium might partially explain the empirical failure of the EH, the scale of the rejection is simply too large to be fully accounted for in this way (see Backus et al., 1989; Dai and Singleton, 2000; Duffee, 2002).

An alternative potential explanation for the rejection of the EH is that there are statistical problems with the tests that are commonly used in the literature. For example, Stambaugh (1988) shows that measurement error in the long yield potentially biases tests of the EH in favor of its rejection. However, Campbell and Shiller (1991) use instrumental variables to show that the EH is strongly rejected even after allowance is made for such measurement error. Bekaert et al. (1997) identify a further small sample bias in tests of the EH, and show using Monte Carlo simulation that even in the relatively large samples that are typically used in empirical work, this bias remains significant.⁴ However, the direction of the bias is such that the empirical evidence actually represents unambiguously *stronger* evidence against the EH than asymptotic theory would imply.⁵

In this paper, we contribute to this debate by examining whether the empirical evidence against the EH is any weaker in data generated since the widespread acceptance in the early 1990s of its failure to explain long yields. If the reported failure of the EH is an asset pricing anomaly, this would imply the existence of potentially profitable trading opportunities in the bond market. As this anomaly became widely known, we might expect market participants to trade so as to move bond yields closer to their equilibrium values required by the EH, and consequently for the evidence against the EH to be weaker in subsequent data. However, if there exists a rational explanation – yet to be uncovered – for the apparent rejection of the EH, there is no reason to expect the nature of this evidence to have systematically changed over time. The majority of studies that have tested the EH (including many of those cited above) employ the monthly synthetic zero-coupon bond yield data of McCulloch (1990), which cover the period December 1946 to February 1987, subsequently updated by McCulloch and Kwon (1993) to February 1991. We extend the zero-coupon bond yield data of McCulloch and Kwon (1993) to December 2009 using data on coupon paying bonds from the CRSP US Treasury Database. We update the empirical evidence on the EH, applying the yield spread tests and VAR tests of Campbell and Shiller (1991) and the forward yield tests of Fama (1984) and Fama and Bliss (1987) to the extended sample as well as to two sub-samples that comprise the McCulloch and Kwon (MK) sample and the additional post-MK sample, respectively.

We find that the evidence against the EH is very much weaker in the post-MK sample than in the original MK sample. In many cases, the EH cannot be rejected in the later data. Moreover, our results are not simply the result of reduced statistical power: across all of the tests that we use, and for almost all of the bond maturities that we consider, the estimated coefficients in the EH tests are substantially closer to unity (their value under the EH) in the post-MK sample than they are in the MK sample. For example, a key prediction of the EH is that the yield spread should forecast subsequent changes in long yields so that the resulting capital gain on the long bond exactly offsets the current spread (except for a possible risk premium). This is a particularly interesting prediction because if it does not hold then there is an immediate profit opportunity. It is rejected at a high significance level for all bond maturities in the MK sample, but in the post-MK sample it is not rejected for any bond with a maturity longer than 6 months. Our results there-

fore offer new hope for the EH as a description of the relationships between the yields of bonds of different maturities and their co-movement through time.

The outline of this paper is as follows. In Section 2, we summarize the theory of the EH and the empirical tests that have been widely used to test it. Section 3 describes the construction of the new dataset of zero-coupon bond yields that we use in the empirical analysis. In Section 4, we replicate all of the conventional tests of the EH using the extended dataset and the two sub-samples. Section 5 concludes.

2. Theoretical background: The expectations hypothesis

Consider an n -period zero-coupon bond with unit face value, whose price at time t is $P_{n,t}$. The yield to maturity of the bond, $Y_{n,t}$, satisfies the relation

$$P_{n,t} = \frac{1}{(1 + Y_{n,t})^n} \quad (1)$$

or, in natural logarithms,

$$p_{n,t} = -ny_{n,t} \quad (2)$$

where $p_{n,t} = \ln(P_{n,t})$ and $y_{n,t} = \ln(1 + Y_{n,t})$. If the bond is sold before maturity then the log m -period holding period return, $r_{n,t+m}^m$, where $m < n$, is defined as the change in log price, $p_{n-m,t+m} - p_{n,t}$, which using (2) can be written as

$$r_{n,t+m}^m = p_{n-m,t+m} - p_{n,t} = ny_{n,t} - (n-m)y_{n-m,t+m} \quad (3)$$

The expectations hypothesis states that the expected holding period return for bonds of different maturities should be equal, except for a risk premium. Combined with the rational expectations hypothesis, the expectations hypothesis of the term structure has a number of important implications for the relationships between bond yields and their movement over time. In particular, the expectations hypothesis states that the expected n -period return on an investment in a series of one-period bonds should be equal to the (certain) n -period return on an n -period bond, which implies that the n -period long yield should be an average of the expected short yield over the following n periods, plus a constant risk premium. That is

$$y_{n,t} = \frac{1}{n} \sum_{i=0}^{n-1} E_t(y_{1,t+i}) + \phi_n \quad (4)$$

where ϕ_n is the risk premium and $E_t(\cdot)$ is the expectation conditional on the time t information set. The relation given by (4) is known as the expectations hypothesis (EH).

The most well known tests of the EH are those of Campbell and Shiller (1991). These tests focus on the predictive ability of the (log) yield spread between long maturity and short maturity bonds, defined as $s_{n,t} = y_{n,t} - y_{1,t}$. In particular, combining Eq. (4) for two adjacent bond maturities and then rearranging gives

$$E_t y_{n-1,t+1} - y_{n,t} = \frac{1}{n-1} (y_{n,t} - y_{1,t}) + \phi_{n-1} - \frac{n}{n-1} \phi_n \quad (5)$$

which states that the yield spread should predict the following period's expected change in the yield on the long bond. Alternatively, rearranging Eq. (4) gives

$$\sum_{i=1}^{n-1} \frac{E_t y_{1,t+i}}{n-1} - y_{1,t} = \frac{n}{n-1} (y_{n,t} - y_{1,t}) + \phi_n \quad (6)$$

which states that the yield spread should predict the cumulative expected change in the short yield over the life of the long bond. These two predictions of the EH can be tested with regressions of the form

⁴ This bias is related to the downward bias of the OLS estimator of the autoregressive coefficient in the short yield model (see Kendall, 1954).

⁵ For further discussion of the statistical properties of tests of the EH, see also Bekaert and Hodrick (2001), Kool and Thornton (2004) and Thornton (2006).

$$y_{n-1,t+1} - y_{n,t} = \alpha_1 + \beta_1 \frac{1}{n-1} (y_{n,t} - y_{1,t}) + \varepsilon_{1,t+1} \quad (7)$$

$$\sum_{i=1}^{n-1} \frac{y_{1,t+i} - y_{1,t}}{n-1} = \alpha_2 + \beta_2 \frac{n}{n-1} (y_{n,t} - y_{1,t}) + \varepsilon_{2,t+1} \quad (8)$$

If the EH holds then the coefficients β_1 and β_2 should be equal to unity, while the intercepts α_1 and α_2 capture the constant risk premium terms. Estimating regression (7) generates a very significant rejection of the EH. The coefficient β_1 is typically found to be significantly less than unity and falls with the maturity of the long bond. For long maturity bonds, it is significantly less than zero. The coefficient β_2 in Eq. (8), in contrast, is typically found to be significantly less than unity for short maturity bonds, but it rises with maturity. For very long maturity bonds, it is often found to be significantly greater than unity (see, for example, Campbell and Shiller, 1991; Bekaert et al., 1997; Bekaert and Hodrick, 2001).⁶ The fact that regression (7) delivers a significant rejection of the EH for all bond maturities but regression (8) does not is ostensibly puzzling (see, for example, Campbell, 1995). However, Bekaert et al. (1997) show that while both regression (7) and regression (8) are subject to small sample biases, the bias is much greater for regression (8) than it is for regression (7). Once this small sample bias is allowed for, regression (8) also delivers a decisive rejection of the EH.

Campbell and Shiller (1991) also propose a vector autoregression (VAR) approach, based on Campbell and Shiller (1987). In particular, a p th-order VAR for the n -period spread, $s_{n,t}$, and the change in the short yield, $\Delta y_{1,t}$, can be written in companion form as

$$Z_{n,t} = AZ_{n,t-1} + \varepsilon_{4,t} \quad (9)$$

where $Z_{n,t}$ is a $(2p \times 1)$ vector comprising the current value and $p-1$ lags of $s_{n,t}$ and the current value and $p-1$ lags of $\Delta y_{1,t}$, A is a $(2p \times 2p)$ matrix of parameters and $\varepsilon_{4,t}$ is a $(2p \times 1)$ vector of errors. Forecasts of the n -period spread and the change in the short yield are then given by $\hat{Z}_{n,t+i} = A^i Z_{n,t}$. Using the EH relation (4), we can then define the 'theoretical' spread as

$$\tilde{s}_{n,t} = e' A [I - (1/n)(I - A^n)(I - A)^{-1}] (1 - A)^{-1} Z_{n,t} \quad (10)$$

where e is a $(1 \times 2p)$ 'selection' vector, such that $e' Z_{n,t} = s_{n,t}$ and I is the $(2p \times 2p)$ identity matrix. Since the conditioning information in the VAR includes the current n -period spread, which itself embodies the market's expectations of future short yields over the life of the long bond, the theoretical spread should be equal to the actual spread. Campbell and Shiller (1991) suggest the following two tests of the EH. Firstly, the correlation between the theoretical spread and the actual spread should be equal to unity. Secondly, the ratio of the standard deviation of the theoretical spread to the standard deviation of the actual spread should be equal to unity. Using the McCulloch (1990) dataset, Campbell and Shiller (1991) find that while the correlation coefficient is indeed close to unity, the standard deviation ratio is typically around 0.5, thus strongly rejecting the EH.

A final way to test the EH focuses on the predictive ability of the expectations of future spot yields that are implicit in the term structure of interest rates. By combining expression (4) for bonds of two different maturities, we can define the m -period forward yield for an n -period bond as

$$f_{n,t}^m = ((n+m)y_{n+m,t} - my_{m,t})/n \\ = E_t(y_{n,t+m}) + ((n+m)\phi_{n+m,t} - m\phi_{m,t})/n \quad (11)$$

Early tests of the EH directly examined whether the forward rates that are implied by the term structure are unbiased predictors of future interest rates. This can be tested using a regression of the form

$$y_{n,t+m} - y_{n,t} = \alpha_3 + \beta_3 (f_{n,t}^m - y_{n,t}) + \varepsilon_{3,t+m} \quad (12)$$

If forward rates are unbiased then the slope coefficient, β_3 , should be equal to unity, while the constant risk premium differential is captured by the intercept, α_3 . This regression has been estimated for values of m between 1 month and 20 years, and for values of n between 1 month and 5 years. While forward yields clearly contain information that is relevant for future spot yields, the estimated coefficient, β_3 , is usually found to be significantly less than unity (see, for example, Fama, 1984, 2006; Fama and Bliss, 1987).

3. Zero-coupon bond yield data

In this paper, we use monthly zero-coupon bond yields on US Treasury securities for the period January 1952 to December 2009.⁷ Many of the empirical studies of the EH described in the preceding section employ the McCulloch (1990) monthly US term structure data, or the subsequently extended data of McCulloch and Kwon (1993).⁸ The McCulloch and Kwon (1993) data comprise monthly time series of estimated zero-coupon yields, par bond yields and instantaneous forward rates (and their respective standard errors) from December 1946 to February 1991. The data are continuously compounded and recorded as annual percentages. Synthetic zero-coupon bond yields are available for 56 maturities from overnight to 40 years.

For the purpose of this paper, we have updated the McCulloch and Kwon (MK) data to December 2009. The data are constructed using the tax-adjusted cubic spline method of McCulloch (1975). The raw data were obtained from the CRSP US Treasury Database and include all available quotations on US Treasury bills, notes and bonds. Data on tax rates were obtained from the Internal Revenue Service, US Department of Treasury.⁹ Since the raw data that we use originate from a different source, it is important to check the integrity of the resulting estimated zero-coupon bond yields. We therefore computed zero-coupon bond yields over a 6-year overlapping period, August 1985 to February 1991, and compared these with the corresponding yields reported in the MK data set.¹⁰ Panel A of Table 1 reports summary statistics for the two data sets for the 10 bond maturities that we use in the empirical analysis. For all 10 maturities, the correlation between the two data sets is in excess of 0.99, and for all except the 1-month maturity, the correlation is in excess of 0.999.

Fig. 1 plots the estimated yields for maturities of 1 month, 1 year and 10 years over the overlapping period. For the 1-year and 10-year maturities, there is no discernable difference between the two data sets. For the 1-month maturity, there are some very minor discrepancies that arise mainly from small differences in the sample of bonds used in the estimation procedure. Panel B of Table 1 reports summary statistics for all ten maturities for the period covered by the MK data (January 1952 to February 1991), for the post-MK data (March 1991 to December 2009), and for the combined sample.

⁷ Although data are available from December 1946, the quality of the estimated data improves significantly after the Treasury Accord of 1951 and so only data after this period are used, as recommended by McCulloch and Kwon (1993).

⁸ The McCulloch and Kwon (1993) zero-coupon bond yield dataset is available at www.econ.ohio-state.edu/jhm/ts/mccckwon/mccull.htm.

⁹ www.irs.gov.

¹⁰ We choose the start date of August 1985 for the overlapping period on the grounds of convenience. Before this date, there are many more irregular bonds in the raw data which have to be manually deleted.

⁶ Campbell and Shiller (1991) also test the EH using analogous regressions based on the yield spread between all possible pairs of bond maturities, $s_{n,t} = y_{n,t} - y_{m,t}$, for n between 2 months and 120 months and for m between 1 month and 60 months. The EH is strongly rejected for almost all pairs of bonds.

Table 1
Summary statistics.

<i>n</i> (months)	McCulloch–Kwon data (August 1985–February 1991)				New data (August 1985–February 1991)				Correlation			
	Mean	Std. error	Min.	Max.	Mean	Std. error	Min.	Max.				
<i>Panel A</i>												
1	6.528	1.193	3.800	9.043	6.527	1.176	3.948	9.248	0.99125			
3	6.936	1.057	5.242	9.053	6.935	1.058	5.243	9.056	0.99963			
6	7.104	0.977	5.262	9.279	7.102	0.976	5.261	9.314	0.99975			
9	7.252	0.947	5.396	9.360	7.247	0.948	5.394	9.362	0.99949			
12	7.388	0.927	5.485	9.490	7.390	0.931	5.488	9.545	0.99953			
24	7.726	0.821	5.988	9.454	7.734	0.824	6.003	9.527	0.99938			
36	7.918	0.775	6.247	9.417	7.915	0.780	6.218	9.470	0.99923			
48	8.046	0.748	6.505	9.559	8.044	0.747	6.535	9.572	0.99967			
60	8.149	0.743	6.648	9.859	8.155	0.738	6.775	9.899	0.99930			
120	8.486	0.715	7.274	10.459	8.479	0.716	7.271	10.459	0.99957			
<i>n</i> (months)	McCulloch–Kwon data (January 1952–February 1991)				New data (March 1991–December 2009)				Extended data (January 1952–December 2009)			
	Mean	Std. error	Min.	Max.	Mean	Std. error	Min.	Max.	Mean	Std. error	Min.	Max.
<i>Panel B</i>												
1	5.314	3.064	0.249	16.210	3.428	1.731	0.021	6.210	4.701	2.843	0.021	16.210
3	5.640	3.143	0.615	15.999	3.610	1.784	0.041	6.291	4.981	2.932	0.041	15.999
6	5.884	3.178	0.685	16.511	3.737	1.784	0.138	6.456	5.186	2.977	0.138	16.511
9	6.003	3.182	0.766	16.635	3.847	1.783	0.209	6.876	5.303	2.980	0.209	16.635
12	6.079	3.168	0.847	16.345	3.949	1.785	0.272	7.142	5.387	2.966	0.272	16.345
24	6.272	3.124	1.149	16.145	4.261	1.730	0.663	7.569	5.619	2.905	0.663	16.145
36	6.386	3.087	1.412	15.825	4.519	1.630	1.007	7.684	5.780	2.839	1.007	15.825
48	6.467	3.069	1.595	15.847	4.749	1.554	1.169	7.712	5.909	2.790	1.169	15.847
60	6.531	3.056	1.770	15.696	4.917	1.471	1.502	7.911	6.007	2.752	1.502	15.696
120	6.683	3.013	2.341	15.065	5.515	1.235	2.792	8.325	6.303	2.630	2.341	15.065

Notes: The table reports summary statistics for the [McCulloch and Kwon \(1993\)](#) and new zero-coupon bond yield datasets for the ten bond maturities that are used in the paper. Panel A reports summary statistics for the overlapping period 08/1985–02/1991. Panel B reports summary statistics for the two sub-samples 01/1952–02/1991 and 03/1991–12/2009, and for the full sample.

[Fig. 2](#) plots the estimated yields for maturities of 1 month, 1 year and 10 years over the full sample. A striking feature of the extended dataset is the evident structural break in each of the time series around 1980–82. Prior to this, yields secularly increased, while after this, they secularly decreased. This has important implications for empirical tests of the EH. Under the EH, rational expectations of this change in the long term trajectory of interest rates would be impounded in long bond yields and so there is no need to explicitly accommodate the structural break under the null hypothesis that the EH holds. However, under any alternative hypothesis that the EH does not hold (for example in the presence of behaviourally biased expectations), we cannot rule out the possibility that there is a corresponding structural break in agents' expectation errors linked to the break between the long upward and downward trends. Following [Fama \(2006\)](#), we accommodate the structural break by including a dummy variable in the intercept of all of the regression tests, equal to zero before the breakpoint and one after the breakpoint. In order to establish the precise location of the structural break, we employ the structural stability test of [Hansen \(1992\)](#). This indicates a breakpoint between June 1981 and June 1982 depending on the regression estimated. For consistency, we assume a common breakpoint at December 1981.¹¹

¹¹ We tested the robustness of the analysis to the location of the breakpoint. In general, we found that the empirical results are not sensitive to the exact choice of breakpoint. We also applied all of the tests of the EH to the full sample excluding the dummy variable. While the dummy variable is highly significant when it is included, its exclusion has very little impact on the evidence in relation to the EH. Results for the analysis excluding the dummy variable are available on request.

4. Results

In this section, we report the results of the regression and VAR tests of the EH, applied to (i) the full sample, January 1952 to December 2009, (ii) the MK sample, January 1952 to February 1991, and (iii) the post-MK sample, March 1991 to December 2009. The single equation regressions for the forward yield, short yield and long yield, and the two equations of the VAR for the short yield and the yield spread, are estimated by OLS. For all of the regressions we estimate heteroscedasticity adjusted standard errors of the parameter estimates and for the regressions in which the dependent variable is overlapping we further allow for serial correlation of a lag order equal to the number of overlapping observations.

4.1. Yield spread tests

[Table 2](#) reports the estimated parameters from the long yield regression (7) for the full sample and the MK and post-MK samples. Standard errors are reported in parentheses. The regressions are estimated for long bonds of maturity $n = 3, 6, 9, 12, 24, 36, 48, 60$ and 120 months. For both the full sample ([Panel A](#)) and the MK sample ([Panel B](#)) the estimated slope coefficient is negative and significantly lower than unity for all bond maturities, and the point estimate falls monotonically with maturity. For all but the 3-month maturity the estimated slope coefficient is not only significantly less than unity, but also significantly less than zero, implying that long bond yields move in a direction opposite to that implied by the EH. The results for the MK sample are similar to those reported by [Campbell and Shiller \(1991\)](#) and [Bekaert et al. \(1997\)](#) who use the same sample. The results for the post-MK sample are reported in [Panel C](#). The EH is only rejected in the post-MK sample for the 3-month and 6-month maturities, which is in sharp

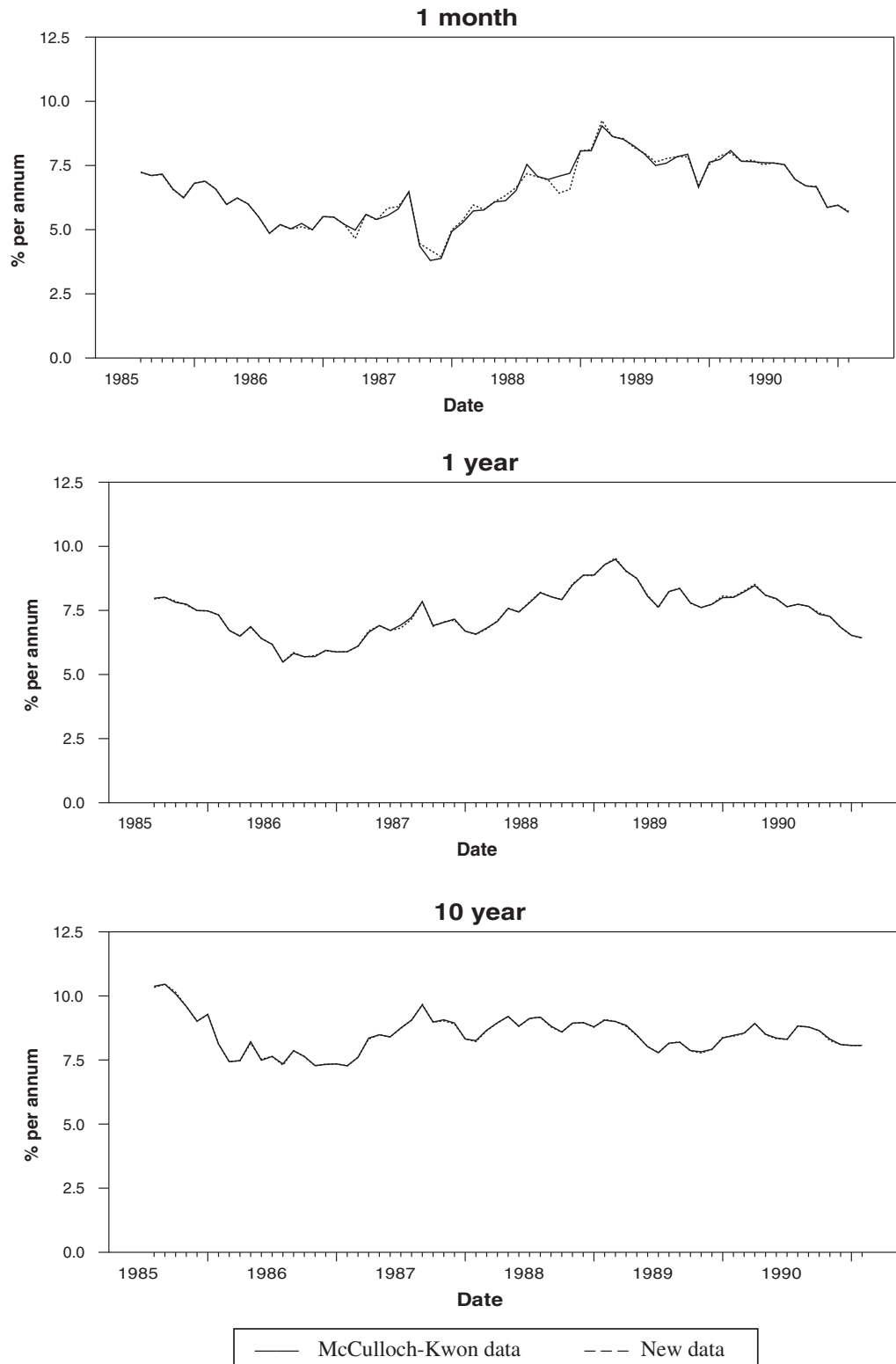


Fig. 1. McCulloch-Kwon and new zero-coupon bond yields for 08/1985–02/1991. The figure plots the [McCulloch and Kwon \(1993\)](#) bond yields and the bond yields constructed in this paper over the overlapping period 08/1985–02/1991 for three bond maturities.

contrast with the MK sample where the EH is rejected for all maturities. The estimated slope coefficient in the post-MK sample falls with bond maturity, becoming negative at the 48-month maturity but, unlike the earlier sample, never significantly so. For all maturities the point estimate of the coefficient on the yield spread is closer to unity in the post-MK sample than it is in the MK sample.

[Table 3](#) reports the estimated parameters from the short yield regression (8). Standard errors (reported in parentheses) are estimated using the [Newey and West \(1987\)](#) estimator to allow for the fact that the dependent variable is overlapping. The regressions are estimated for long bonds of maturity $n = 3, 6, 9, 12, 24, 36, 48, 60$ and 120 months. In the post-MK sample the EH is only rejected

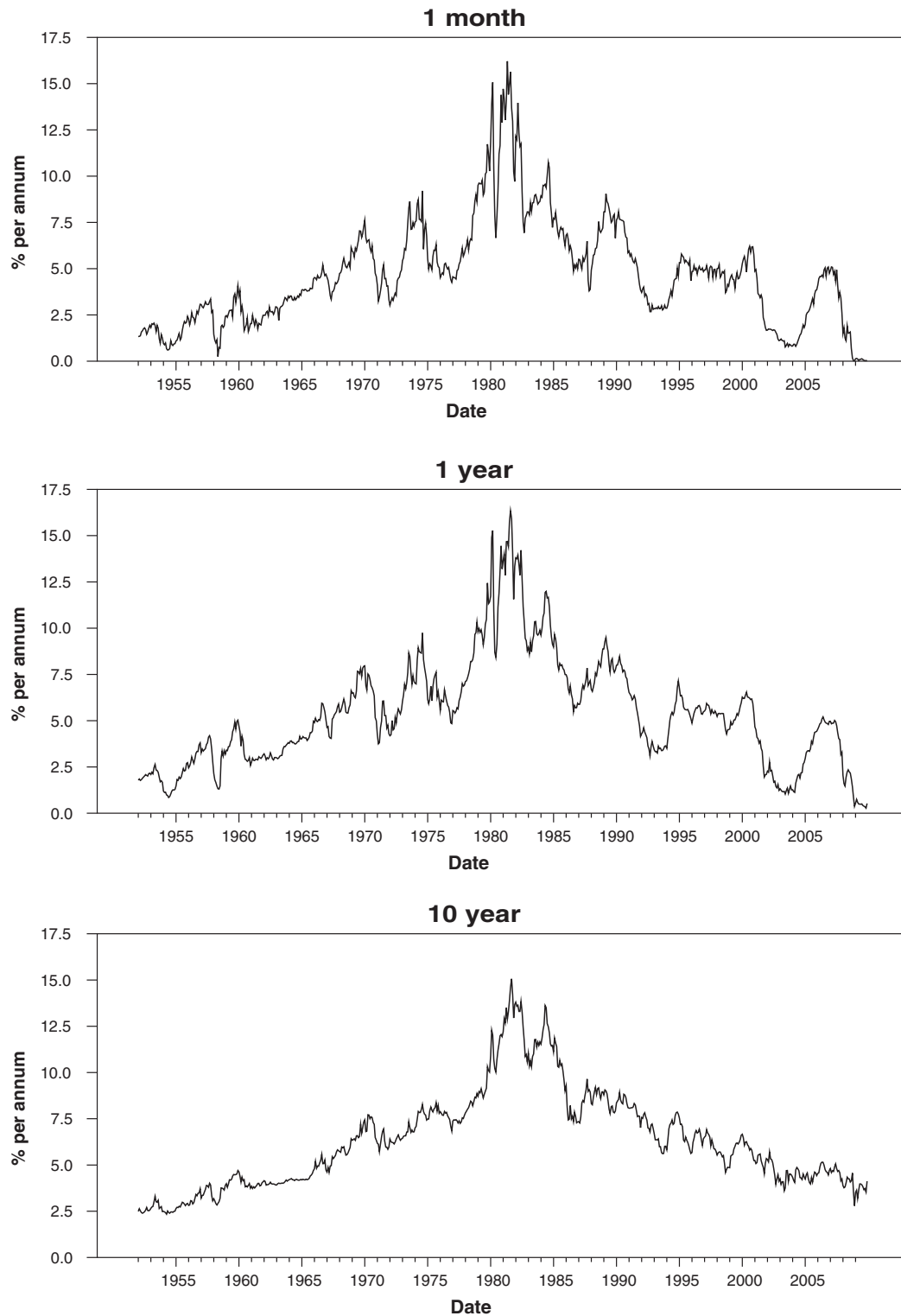


Fig. 2. Zero-coupon bond yields for the full sample 01/1952–12/2009 for three bond maturities.

for the 3-month maturity and (marginally) the 120-month maturity. This contrasts with results for both the full sample and the MK sample, where the EH is rejected for all short maturity bonds up to 24 months and for the 120-month bond. The results that we report for the MK sample are again consistent with those reported by [Campbell and Shiller \(1991\)](#) and [Bekaert et al. \(1997\)](#). For all but one of the maturities the point estimate of the slope

coefficient is closer to unity in the post-MK sample than in the MK sample.

4.2. VAR tests

[Table 4](#) reports the correlation coefficient and the standard deviation ratio between the actual yield spread and the theoretical

Table 2
Long yield regression.

<i>n</i>	Panel A: Full sample			Panel B: MK sub-sample			Panel C: Post-MK sub-sample		
	α_1	γ_1	β_1	α_1	γ_1	β_1	α_1	γ_1	β_1
3	−0.071 (0.033)	−0.050 (0.038)	−0.115 (0.134)	−0.078 (0.041)	−0.127 (0.065)	−0.068 (0.183)	−0.106 (0.021)	–	0.128 (0.151)
6	0.021 (0.035)	−0.054 (0.037)	−0.558 (0.222)	0.045 (0.044)	−0.046 (0.063)	−0.790 (0.307)	−0.094 (0.021)	–	0.451 (0.242)
9	0.067 (0.035)	−0.084 (0.037)	−0.807 (0.300)	0.091 (0.044)	−0.070 (0.064)	−1.117 (0.417)	−0.078 (0.024)	–	0.338 (0.344)
12	0.070 (0.033)	−0.088 (0.036)	−0.909 (0.357)	0.091 (0.042)	−0.075 (0.063)	−1.244 (0.497)	−0.075 (0.027)	–	0.335 (0.439)
24	0.056 (0.029)	−0.052 (0.034)	−0.792 (0.518)	0.072 (0.035)	−0.033 (0.058)	−1.279 (0.714)	−0.046 (0.033)	–	0.548 (0.735)
36	0.058 (0.026)	−0.046 (0.032)	−1.179 (0.634)	0.070 (0.031)	−0.032 (0.053)	−1.664 (0.852)	−0.026 (0.036)	–	0.033 (0.957)
48	0.058 (0.025)	−0.041 (0.030)	−1.496 (0.730)	0.068 (0.029)	−0.032 (0.049)	−1.988 (0.973)	−0.013 (0.037)	–	−0.401 (1.099)
60	0.057 (0.023)	−0.038 (0.029)	−1.720 (0.806)	0.066 (0.027)	−0.033 (0.046)	−2.281 (1.070)	−0.009 (0.037)	–	−0.545 (1.217)
120	0.053 (0.019)	−0.031 (0.025)	−2.663 (1.128)	0.062 (0.021)	−0.035 (0.037)	−3.600 (1.518)	0.003 (0.034)	–	−1.165 (1.645)

Notes: The table reports the results of estimating the long yield regression (7) in the main text. The regressions include a dummy variable equal to one after December 1981 and zero otherwise. The parameter on the dummy variable is denoted γ_1 . Results are reported for the full sample, 01/1952–12/2009 (Panel A), the MK sub-sample, 01/1952–02/1991 (Panel B) and the post-MK sub-sample, 03/1952–12/2009 (Panel C). Standard errors are reported in parentheses.

Table 3
Short yield regression.

<i>n</i>	Panel A: Full sample			Panel B: MK sub-sample			Panel C: Post-MK sub-sample		
	α_2	γ_2	β_2	α_2	γ_2	β_2	α_2	γ_2	β_2
3	−0.114 (0.039)	−0.052 (0.037)	0.488 (0.107)	−0.120 (0.043)	−0.158 (0.060)	0.510 (0.126)	−0.151 (0.021)	–	0.684 (0.074)
6	−0.131 (0.073)	−0.137 (0.084)	0.394 (0.130)	−0.118 (0.074)	−0.291 (0.150)	0.369 (0.149)	−0.330 (0.055)	–	0.856 (0.103)
9	−0.126 (0.096)	−0.245 (0.124)	0.388 (0.131)	−0.090 (0.090)	−0.415 (0.220)	0.331 (0.139)	−0.483 (0.105)	–	0.905 (0.150)
12	−0.152 (0.124)	−0.361 (0.164)	0.452 (0.159)	−0.116 (0.116)	−0.564 (0.273)	0.398 (0.166)	−0.652 (0.165)	–	0.986 (0.189)
24	−0.250 (0.252)	−0.796 (0.331)	0.659 (0.187)	−0.257 (0.257)	−1.233 (0.527)	0.667 (0.195)	−1.090 (0.378)	–	1.054 (0.283)
36	−0.275 (0.331)	−1.193 (0.418)	0.776 (0.192)	−0.315 (0.348)	−1.999 (0.733)	0.822 (0.185)	−1.317 (0.438)	–	1.034 (0.244)
48	−0.340 (0.319)	−1.650 (0.453)	0.930 (0.195)	−0.352 (0.340)	−2.890 (0.792)	0.943 (0.206)	−1.610 (0.541)	–	1.072 (0.254)
60	−0.382 (0.274)	−2.098 (0.453)	1.036 (0.186)	−0.509 (0.275)	−4.254 (0.547)	1.168 (0.166)	−2.007 (0.513)	–	1.148 (0.248)
120	−0.211 (0.364)	−3.690 (0.534)	1.134 (0.167)	−0.352 (0.317)	–	1.402 (0.153)	−2.545 (0.130)	–	0.850 (0.068)

Notes: The table reports the results of estimating the short yield regression (8) in the main text. The regressions include a dummy variable equal to one after December 1981 and zero otherwise. The parameter on the dummy variable is denoted γ_2 . Results are reported for the full sample, 01/1952–12/2009 (Panel A), the MK sub-sample, 01/1952–02/1991 (Panel B) and the post-MK sub-sample, 03/1952–12/2009 (Panel C). Standard errors are reported in parentheses.

yield spread derived from the VAR, for bond maturities $n = 3, 6, 12, 24, 36, 48, 60$ and 120 months. The VAR was specified with a lag length of four chosen on the basis of the Schwartz Bayesian criterion.

The standard deviation test decisively rejects the EH in both the full sample and the MK sample. In particular, the standard deviation ratio is significantly lower than unity for all bond maturities using the full sample and for all but the longest bond maturities using the MK sample. These results are again consistent with the findings of Campbell and Shiller (1991). In contrast, in the post-MK sample the EH is rejected only for the 3-month maturity. For

all maturities the standard deviation ratio is closer to unity in the post-MK sample than it is in the MK sample. In many cases the differences are striking. For example, for the 12-month maturity the standard deviation ratio, which should be one under the EH, is estimated to be 0.943 in the post-MK sample but only 0.462 in the MK sample.

In the case of the correlation coefficient test, for both the full sample and the MK sample the EH cannot be rejected at conventional significance levels owing to the relatively large standard error with which the correlation coefficient is estimated. In the post-MK sample the EH is again not rejected, but the correlation

coefficient is now closer to unity than it is in the MK sample for all bond maturities. Indeed the minimum value of the correlation coefficient is 0.714 in the MK sample but 0.934 in the post-MK sample.

4.3. Forward yield tests

Table 5 reports the estimated parameters from the forward yield regression (12) for the 1-month bond maturity ($n = 1$ month) and forward horizons of between 1 month and 1 year ($m = 1, 3, 6, 9$ and 12 months). In both the full sample and the MK sample the estimated slope coefficient at first declines with maturity and then rises with maturity, but is significantly less than unity for all forward horizons. These results are consistent with the findings of Fama (1984). However, for the post-MK sample we cannot reject the null hypothesis that the slope coefficient is equal to one except for the 1-month forward horizon, and only marginally for the 3-month forward horizon. In all cases the estimated slope coefficient

is substantially closer to unity in the post-MK sample than it is in the MK sample. The average value of the slope coefficient across all five forward horizons is 0.458 in the MK sample but 0.837 in the post-MK sample.

Table 6 reports the results of the same regression for the 1-year bond maturity ($n = 12$ months) and forward horizons of between 1 and 10 years ($m = 12, 24, 36, 48, 60$ and 120 months). In the full sample the EH is rejected for four of the six forward horizons, while in the MK sample it is rejected for five of the six forward horizons. For the full sample and the MK sample the estimated slope coefficient is significantly less than unity for the 12-month horizon, while for the MK sample it is significantly less than unity for the 24-month horizon also. For the 36-month, 48-month and 60-month horizons the slope coefficient is significantly greater than unity. For the 120-month horizon the coefficient is not significantly different from unity for either the full sample or the MK sample. The results for the MK sample are very similar to those reported by Fama and Bliss (1987) (which covers the period June 1953 to

Table 4
VAR correlation coefficient and standard deviation ratio.

n	Panel A: Full sample		Panel B: MK sub-sample		Panel C: Post-MK sub-sample	
	Correlation	SD ratio	Correlation	SD ratio	Correlation	SD ratio
3	0.883 (0.097)	0.555 (0.093)	0.856 (0.122)	0.597 (0.109)	0.952 (0.037)	0.710 (0.068)
6	0.826 (0.163)	0.484 (0.102)	0.722 (0.251)	0.518 (0.113)	0.935 (0.043)	0.902 (0.123)
9	0.848 (0.157)	0.446 (0.101)	0.714 (0.257)	0.463 (0.107)	0.934 (0.048)	0.928 (0.166)
12	0.886 (0.121)	0.456 (0.112)	0.780 (0.205)	0.462 (0.118)	0.943 (0.043)	0.943 (0.202)
24	0.951 (0.066)	0.490 (0.166)	0.903 (0.131)	0.484 (0.190)	0.969 (0.038)	0.795 (0.233)
36	0.975 (0.037)	0.525 (0.191)	0.954 (0.067)	0.534 (0.229)	0.978 (0.036)	0.683 (0.240)
48	0.984 (0.023)	0.553 (0.195)	0.971 (0.041)	0.574 (0.236)	0.983 (0.033)	0.627 (0.241)
60	0.989 (0.016)	0.577 (0.192)	0.980 (0.028)	0.602 (0.229)	0.989 (0.024)	0.618 (0.245)
120	0.996 (0.005)	0.661 (0.168)	0.993 (0.008)	0.699 (0.192)	0.997 (0.009)	0.609 (0.238)

Notes: The table reports the correlation coefficient and standard deviation ratio between the actual yield spread and the theoretical yield spread given by Eq. (10) in the main text. The estimate VAR includes a dummy variable equal to one after December 1981 and zero otherwise. Results are reported for the full sample, 01/1952–12/2009 (Panel A), the MK sub-sample, 01/1952–02/1991 (Panel B) and the post-MK sub-sample, 03/1952–12/2009 (Panel C). Standard errors are reported in parentheses.

Table 5
Forward yield regression (forecasts of 1-month spot rates).

m	Panel A: Full sample			Panel B: MK sub-sample			Panel C: Post-MK sub-sample		
	α_3	γ_3	β_3	α_3	γ_3	β_3	α_3	γ_3	β_3
1	−0.154 (0.033)	−0.039 (0.040)	0.501 (0.050)	−0.164 (0.040)	−0.152 (0.067)	0.528 (0.067)	−0.154 (0.023)	–	0.565 (0.062)
3	−0.198 (0.066)	−0.143 (0.072)	0.434 (0.066)	−0.202 (0.082)	−0.360 (0.123)	0.439 (0.090)	−0.374 (0.043)	–	0.812 (0.081)
6	−0.128 (0.089)	−0.369 (0.096)	0.360 (0.070)	−0.079 (0.109)	−0.606 (0.161)	0.302 (0.093)	−0.633 (0.076)	–	0.829 (0.096)
9	−0.124 (0.094)	−0.658 (0.112)	0.464 (0.071)	−0.090 (0.111)	−0.980 (0.193)	0.422 (0.093)	−0.965 (0.113)	–	0.982 (0.115)
12	−0.191 (0.104)	−0.956 (0.131)	0.598 (0.070)	−0.193 (0.119)	−1.403 (0.229)	0.600 (0.092)	−1.238 (0.146)	–	0.998 (0.119)

Notes: The table reports the results of estimating the forward yield regression (12) in the main text for $n = 1$. The regressions include a dummy variable equal to one after December 1981 and zero otherwise. The parameter on the dummy variable is denoted γ_3 . Results are reported for the full sample, 01/1952–12/2009 (Panel A), the MK sub-sample, 01/1952–02/1991 (Panel B) and the post-MK sub-sample, 03/1952–12/2009 (Panel C). Standard errors are reported in parentheses.

Table 6

Forward yield regression (forecasts of 1-year spot rates).

<i>m</i>	Panel A: Full sample			Panel B: MK sub-sample			Panel C: Post-MK sub-sample		
	α_3	γ_3	β_3	α_3	γ_3	β_3	α_3	γ_3	β_3
12	0.238 (0.087)	−1.015 (0.130)	0.480 (0.097)	0.255 (0.091)	−1.218 (0.206)	0.403 (0.125)	−0.722 (0.134)	–	0.733 (0.153)
24	0.256 (0.116)	−2.077 (0.177)	0.985 (0.093)	0.344 (0.122)	−2.178 (0.291)	0.726 (0.126)	−1.785 (0.192)	–	1.311 (0.133)
36	0.272 (0.111)	−3.441 (0.175)	1.501 (0.076)	0.365 (0.125)	−3.867 (0.308)	1.284 (0.114)	−2.833 (0.171)	–	1.633 (0.087)
48	0.339 (0.107)	−4.170 (0.170)	1.640 (0.068)	0.331 (0.121)	−5.302 (0.317)	1.657 (0.102)	−3.103 (0.165)	–	1.556 (0.077)
60	0.519 (0.115)	−4.512 (0.190)	1.415 (0.070)	0.351 (0.125)	−6.497 (0.361)	1.744 (0.099)	−2.840 (0.192)	–	1.129 (0.080)
120	1.405 (0.144)	−6.309 (0.255)	1.005 (0.089)	1.674 (0.153)	–	0.929 (0.125)	−2.656 (0.243)	–	0.089 (0.092)

Notes: The table reports the results of estimating the forward yield regression (12) in the main text for $n = 12$. The regressions include a dummy variable equal to one after December 1981 and zero otherwise. The parameter on the dummy variable is denoted γ_3 . Results are reported for the full sample, 01/1952–12/2009 (Panel A), the MK sub-sample, 01/1952–02/1991 (Panel B) and the post-MK sub-sample, 03/1952–12/2009 (Panel C). Standard errors are reported in parentheses.

December 1985) and strongly reject the EH. In the post-MK sample the estimated slope coefficient is closer to unity than in the MK sample for the 12-month, 48-month and 60-month horizons. The EH is not rejected for the 12-month horizon and only marginally rejected for the 24-month horizon. For the 36-month, 48-month and 60-month horizons the slope coefficient is significantly greater than unity, while for the 120-month horizon it is significantly less than unity. The results for the 1-year forward yield tests reported here for the full sample and the post-MK sample are consistent with Fama (2006), who extends the analysis of Fama and Bliss (1987) to include the additional period January 1986 to December 2004 and considers forward horizons of 12, 24, 36 and 48 months. For the full sample the estimated coefficients are similar to those reported by Fama for the period June 1953 to December 2004, while for the post-MK sample the estimated coefficients reported here are similar to those reported by Fama for the period January 1986 to December 2004.

5. Conclusion

By the early 1990s there was overwhelming evidence that the EH did not describe how long yields are determined in the market. The focus of most subsequent empirical work on the term structure therefore turned to attempts to explain why the EH does not hold. However, nearly 20 years of data have accumulated since these tests were conducted. The purpose of this paper is to revisit the original tests using this new data. Using data on coupon paying bonds from the CRSP US Treasury Database, we extend the zero-coupon bond yield data of McCulloch and Kwon (1993), which ended in February 1991, to December 2009. We apply a range of tests to short yields, long yields and forward yields using both the extended sample and sub-samples that comprise the McCulloch and Kwon (1993) data and the more recent period, 1991–2009.

We find that the evidence against the EH is very much weaker in the 1991–2009 period than in the original McCulloch and Kwon sample. Indeed in many cases the EH can no longer be rejected in the later sample. For example, the EH predictions for changes in the long yield are decisively rejected for every bond maturity in the McCulloch and Kwon sample but cannot be rejected for any bond with a maturity of more than 6 months in the 1991–2009 period. The estimated coefficients are closer to the value implied by the EH in almost all of the tests that we employ and across almost all bond maturities. A plausible explanation for this finding is the widespread dissemination of evidence rejecting the EH based

on the McCulloch and Kwon data set, with the implication of a simple trading rule to exploit this inefficiency. Expected profits should accrue to a trading rule that is long in long bonds when the long rate is greater than the short rate and short in long bonds when the long rate is less than the short rate.

The implementation of such a trading rule was facilitated by a number of financial innovations in the US, starting in the 1980s, which reduced transactions costs and increased liquidity in the bond market. For example, the introduction in the 1980s of interest rate swaps, which allow an investor to exchange the cash flows of a floating rate instrument for the cash flows of a fixed rate instrument, provided an ideal vehicle for exploiting such anomalies. The development of the Separate Trading of Registered Interest and Principal of Securities (STRIPS) market by investment banks in the early 1980s, and its adoption by the US Treasury in 1985, resulted in an increase in liquidity for Treasury securities, and a reduction in transactions costs, making it substantially cheaper for investors to arbitrage pricing anomalies in the bond market. At the same time that new products were being introduced, there were also innovations in how they could be traded. The creation of GovPX in 1991 through a consortium of the primary dealers and five of the six interdealer brokers presented traders with improved access to information. The advent of electronic communication networks (such as eSpeed in 1999 and BrokerTec in 2000) also contributed to the sharp decline in transaction costs over the last decade. Together, these innovations resulted in a marked increase in volume that is suggestive of increased arbitrage activity. For example, the Treasury STRIPS market grew from \$23 billion in 1985 to \$173 billion in 2009 while the notional value of the interest rate derivatives market grew from \$3.8 trillion in 1991 to \$180 trillion in 2009.

Alongside these new financial products – and the innovations that facilitated their trading – the last two decades have seen a striking expansion in the number and size of hedge funds, which have both the capital and regulatory flexibility to employ such products in order to exploit mis-pricing that is implied by the failure of the EH. It is noteworthy that while sophisticated financial products and the hedge funds that trade in them have been subject to considerable criticism in the late 2000s, the results we report here are consistent with the argument that in fact they have served to facilitate risky arbitrage, reduce mis-pricing and increase market efficiency.

The results reported in this paper provide further evidence that asset pricing anomalies are transitory, and that they tend to be eroded once they are widely disseminated (see Fama, 1998). For

example, the 'size' effect in equity returns was well-documented by a number of important papers published in the early 1980s (for example Banz, 1981). However, following the publication of these papers – and the emergence of this anomaly into the public domain – the size effect has been greatly diminished (see, for example, Schwert, 2003). The evidence that we report in this paper shows that long yields have moved substantially closer to the rates implied the EH as market participants simultaneously gained information about a profit opportunity, capital through the expansion of hedge funds, and the means to exploit the profit opportunity using new products that entailed lower transactions costs.

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Appendix A

This appendix gives a detailed description of the construction of the synthetic zero-coupon bond yield data that are used in the empirical analysis. The data are constructed using the tax-adjusted cubic spline method employed by McCulloch (1990) and McCulloch and Kwon (1993), which is based on the method developed by McCulloch (1975) but modified by the assumption that bond coupons arrive discretely rather than in a continuous stream. The raw data are taken from the US Treasury Database of the Centre for Research in Security Prices (CRSP), and include all month-end quotations on individual US Treasury bills, notes and bonds. Data on tax rates were obtained from the Internal Revenue Service. Since our source of bond data is different from McCulloch and Kwon's, we check the reliability of the new data by computing yields over a 6-year overlapping period, August 1985 to February 1991, and compare these with the corresponding figures in the McCulloch and Kwon dataset. We applied the same filters as McCulloch and Kwon to select securities to be used in the construction of the term structure.¹² In particular, we removed all tax-exempt bonds and, from 1985 onwards, all callable bonds. There were 26 callable bonds during the overlapping August 1985 to February 1991, and the figure remained at 26 until January 1992. By December 2004 there were only 10 callable bonds and there were none by November 2009. There are no tax-exempt bonds after the mid-1960s. We also removed a number of flower bonds. Flower (or estate tax) bonds are bonds that are redeemable at par in payment of estate taxes if owned by the deceased at the time of death. These bonds had relatively low coupons and should consequently trade at a price significantly lower than par value. However, in practice flower bonds carry substantial premiums. In August 1985, for example, the three non-callable flower bonds (the 3.5% February 1990, the 3.0% February 1995 and the 3.5% November 1998) were trading at about \$90 each, but their coupons and principal were worth only \$77, \$56 and \$49, respectively. They therefore carried premiums of about \$13, \$35 and \$41, respectively. Of the six flower bonds available in August 1985, three were removed on the basis that they were callable. This left only three further flower bonds, which were removed.

CRSP provides quotes for each security on a price basis, and so for Treasury bills, bid/ask discount yields were backed out using

the usual conversion formula, $p = 100 - y * d/360$, where p is the bid/ask price, y is the bid/ask yield and d is the number of days to maturity. For Treasury bonds, the prices were used directly. Following McCulloch and Kwon, we do not take into account broker fees or any other transaction costs in the calculation of zero-coupon yields. After eliminating all tax-exempt, callable and flower bonds, the number of securities each month very closely matches the number in the McCulloch and Kwon data set during the overlapping period August 1985 to February 1991. Indeed, there was only one additional bond in McCulloch and Kwon's original input file for August 1985 and for February 1991 our input file had four additional bonds. As shown in Fig. 1 in the main body of the paper the differences in the computed zero-coupon bond yields are negligible.

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